How-To Library

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Moving DB2 for z/OS Bulk Data with Nonrelational Source Definitions

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Abstract

You can define a DB2 for z/OS data source as a nonrelational data source in a PowerExchange bulk data movement session. This article describes the types of DB2 data sources that you define as nonrelational and explains how to create a data map for the data source and create a bulk data movement session to materialize a target from the data source.

Supported Versions

- PowerExchange 10.0 or later

Table of Contents

Overview. 2
DB2 Unload Files as Bulk Data Sources. 2
Map the DB2 Table. 3
Edit the Record Layout in a DB2 Data Map. 4
Map the DB2 Unload File. 4
Materialize the Target from the DB2 Nonrelational Data Source. 6

Overview

You can define a DB2 for z/OS data source as a nonrelational source. Use this approach with the following DB2 for z/OS data sources:

- DB2 unload files
- DB2 tables that contain columns that you want to split into multiple fields
- DB2 tables for which you want to define fields that use one or more PowerExchange functions in an expression to process source data

To move DB2 for z/OS bulk data that you define as a nonrelational source, complete the following steps:

- Create the unload file, if applicable.
- Create a data map in the PowerExchange Navigator, send the data map to the PowerExchange Listener on the remote system, and run a database row test.
- Materialize the target. In PowerCenter, create a mapping, import the data map, and create and run the workflow.

DB2 Unload Files as Bulk Data Sources

You can use a DB2 unload file as a data source for a PowerExchange bulk data movement.

To create an unload file, you can use any of the following IBM or BMC Software utilities:

- DB2 for z/OS online REORG TABLESPACE utility with UNLOAD EXTERNAL
- DB2 for z/OS online UNLOAD utility
- DB2 for z/OS sample DSNTIAUL unload program
- BMC UNLOAD PLUS for DB2

Note: PowerExchange cannot read large format sequential data sets. If you specify DSNTYPE=LARGE in the DD statement when you create the unload file, PowerExchange cannot read the file. Instead, you can specify DSNTYPE=EXTREQ in the DD statement.

Map the DB2 Table

Use this procedure to create a PowerExchange data map for a DB2 table, send the data map to the PowerExchange Listener on the remote system, and run a database row test.

Before you begin, gather the following information:

- DB2 subsystem ID
- DB2 table names in the format *creator.table_name*
- MVS user ID and password if required by the security settings in the SECURITY statement in the DBMOVER configuration member

1. On the **Resources** tab in the **Resource Explorer**, click **Add > Data Map**.
2. Enter the following information in the **Name** dialog box:
   - Select the **DB2** access method.
   - Select the **Import Record Definitions** option.
3. Click **Next**.
4. In the **DB2 Access Method** dialog box, enter the DB2 instance and table name and click **Finish**.
5. In the **Import Copybook - Source Details** dialog box, select **Remote**.

   In the **Type** list, select the **DB2 Catalog** copybook type.
6. Click **Next**.
7. In the **Import Copybook - Remote DB2 Catalog Details** dialog box, enter the following information:
   - Optionally, change the import details. The input fields are populated from previous selections.
   - In the **Save File Locally As** box, enter a file name.
   - To select specific columns, click **Preview/Amend Columns**.
8. **Click Next**.
9. In the **Import Copybook - Configuration Details** dialog box, select the actions for imported records, fields, and tables and click **Next**.
10. In the **Import Copybook Information** window, review the information for the import and click **OK**.
11. In the **Record Definition** dialog box, select an action for each imported record and click **OK**.

    The **DB2 Catalog Import** window displays the imported copybook.

    The **Copybook Message Log** window displays the results of the import operation.

    Note: If an error message appears in the **Copybook Message Log** window, double-click the message to find the associated line in the copybook.
12. Edit the record layout in the data map as needed. For more information, see "Edit the Record Layout in a DB2 Data Map" on page 4.
13. Define any fields that use PowerExchange functions in an expression to process source data. For more information, see the *PowerExchange Navigator User Guide*.
14. Open the data map in the Resource Explorer, and click the "record" or "table" view to verify that the metadata was successfully retrieved.

15. Send the data map to the PowerExchange Listener on the remote DB2 for z/OS system. In the Resource Explorer, select the data map and click **File > Send to Remote Node**. This step enables PowerExchange extraction routines to access the data map at runtime.

    Note: If you do not complete this step, you are prompted to send the data map to the remote node when you run a database row test.

16. Run a database row test on the table view of the metadata to verify that data can be returned from the DB2 source database.

Edit the Record Layout in a DB2 Data Map

You can edit the record layout of a DB2 table in a DB2 data map.

For example, you can edit a field in a record that contains three packed items to separate the data in the field into child fields.

In this example, the MISC_1 field is defined as a PACKED field with a precision of 9 and a scale of 2.

To edit the record layout in a DB2 data map:

1. Open the data map.
2. Open the record.
3. In the **Record** window, right-click the MISC_1 field and click **Properties**.

    The **Field Properties** dialog box appears.

4. In the **Field Properties** dialog box, enter the following information:
    - In the **Field Name** box, enter `MISC_DATA`.
    - In the **Field Type**, enter `GROUP`.
5. Click **OK**.
6. In the **Record** window, right-click the MISC_DATA field and click **Add Field as Child**.
7. Add the MISC_1, MISC_2, and MISC_3 fields as children of the MISC_DATA field. Define each field as a PACKED field with a precision of 9 and a scale of 2.

Map the DB2 Unload File

Use this procedure to create a PowerExchange data map for a DB2 unload file, send the data map to the PowerExchange Listener on the remote system, and run a database row test.

Before you begin, create the DB2 unload file. In addition, gather the following information:

- DB2 subsystem ID
- DB2 table names in the format *creator.table_name*
- MVS user ID and password if required by the security settings in the SECURITY statement in the DBMOVER configuration member
- Name of the unload file

1. On the **Resources** tab in the **Resource Explorer**, click **Add > Data Map**.
2. Enter the following information in the **Name** dialog box:
    - Select the **DB2UNLD** access method.
    - Select the **Import Record Definitions** option.

3. In the **DB2UNLD Access Method** dialog box, enter the following information:
   - In the **File Name** box, enter the name of the DB2 unload file.
   - In the **DB Instance** box, enter the DB instance, such as DSN9.
   - In the **Table Name** box, enter the table name or click the Browse button to browse to the file. In the **DB2 Table Filter** dialog box, enter filter criteria based on the creator or schema name, database name, or table name. Click **Next**. Select a table and click **Finish**.
   - In the **Unload Type** list, select one of the following unload types:
     - REORG UNLOAD
     - DSNTIAUL/BMC UNLOAD+
     - DSNUTILB UNLOAD
     - UNDEFINED

     By default, the UNDEFINED type is generated with null indicators and count fields in the same position as in the REORG UNLOAD EXTERNAL type. You can move or delete these to correspond to user-defined unload formats.
4. Click **Finish**.
5. In the **Import Copybook - Source Details** dialog box, select **Remote** and select the **DB2 Catalog** copybook type.
6. Click **Next**.
7. In the **Import Copybook - Remote DB2 Catalog Details** dialog box, enter the following information:
   - In the **Save File Locally As** box, enter a file name.
   - In the **Null Indicator** box, enter the hexadecimal value of the null indicator that is used in the unload file that you are mapping.
   - If the VARCHAR columns are padded in the unload file that you are mapping, select the **Pad Variable** option. This option corresponds to the BMC UNLOAD PLUS FIXEDVARCHAR option.
   - To select specific columns, click **Preview/Amend Columns**.
8. Click **Next**.
9. In the **Import Copybook - Configuration Details** dialog box, select the actions for imported records, fields, and tables and click **Next**.
10. In the **Import Copybook Information** window, review the information for the import and click **OK**.
11. In the **Record Definition** dialog box, select an action for each imported record and click **OK**.

    The **DB2 Catalog Import** window displays the imported copybook.

    The **Copybook Message Log** window displays the results of the import operation.

    Note: If an error message appears in the **Copybook Message Log** window, double-click the message to find the associated line in the copybook.
12. Open the data map in the Resource Explorer, and click the “record” or “table” view to verify that the metadata was successfully retrieved.
13. Send the data map to the PowerExchange Listener on the remote DB2 for z/OS system. In the Resource Explorer, select the data map and click **File > Send to Remote Node**. This step enables PowerExchange extraction routines to access the data map at runtime.

    Note: If you do not complete this step, you are prompted to send the data map to the remote node when you run a database row test.
14. Run a database row test on the table view of the metadata to verify that data can be returned from the DB2 source database.

Materialize the Target from the DB2 Nonrelational Data Source

Use this procedure to materialize a target from a DB2 nonrelational data source. The procedure assumes that you are using PWXPC to integrate PowerExchange with PowerCenter.

Before you begin, create a data map in PowerExchange for the data source, and run a database row test.

1. In the PowerCenter Designer, import the PowerExchange data map and create a mapping. At **Source Type**, select **DB2UNLD** for a DB2 unload data map or **DB2MAP** for a DB2 table for which you defined a data map.

   In the **Location** field, enter the PowerExchange Listener node name that you specified in the NODE statement of the local dbmover.cfg file. Complete the other fields, as needed.
2. In the PowerCenter Workflow Manager, define a workflow and session and configure a PWX NRDB Batch application connection. Depending on the source type, PowerCenter selects one of the following readers:

   - PowerExchange Reader for DB2 Datamaps
   - PowerExchange Reader for DB2 Unload Datasets

   Then start the workflow to perform the bulk data movement.

Author

Jim Middleton